

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 04 July 2026

Team ID: Individual

Project Name: India's Agricultural Crop Production Analysis (1997-2021)

Maximum Marks: 4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form (Name, Organization, Role) Registration through Gmail Registration as Farmer / Researcher / Admin role
FR-2	User Authentication & Confirmation	Confirmation via Email verification link Login via Email/Username & Password Confirmation via OTP for password reset
FR-3	Data Upload & Management	Upload agricultural crop production dataset (CSV/Excel, 1997-2021) View, edit, or delete uploaded records Validate data format before processing
FR-4	Data Visualization & Analysis (Tableau)	State wise Crop Production analysis Year wise Crop Production trend analysis (1997-2021) Season wise (Kharif / Rabi / Whole Year) Crop Production analysis Crop wise Production and Area comparison Yield analysis (Production vs Area Cultivated) Impact of Rainfall on Crop Production District wise Crop Production analysis Top crop-producing States and Districts Crop diversification trends over the years Comparative analysis of Production across different crop types
FR-5	Report Generation	Generate summary reports of the analysis Export dashboards/reports as PDF or Image
FR-6	Admin Dashboard	View all uploaded datasets and reports Manage user access and permissions

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should be simple and intuitive so that non-technical users such as farmers and policymakers can easily interpret crop production trends.
NFR-2	Security	Access to uploaded agricultural datasets should be restricted through authentication and role-based permissions to protect data integrity.
NFR-3	Reliability	Dashboards should consistently reflect accurate, up-to-date crop production data without errors or data loss.
NFR-4	Performance	Visualizations and dashboards should load quickly even when analyzing 25 years (1997-2021) of large-scale crop production data.
NFR-5	Availability	The dashboard and reports should be accessible to users whenever needed, with minimal downtime.
NFR-6	Scalability	The system should be able to handle growing datasets (additional years, states, or crops) and an increasing number of users without performance degradation.